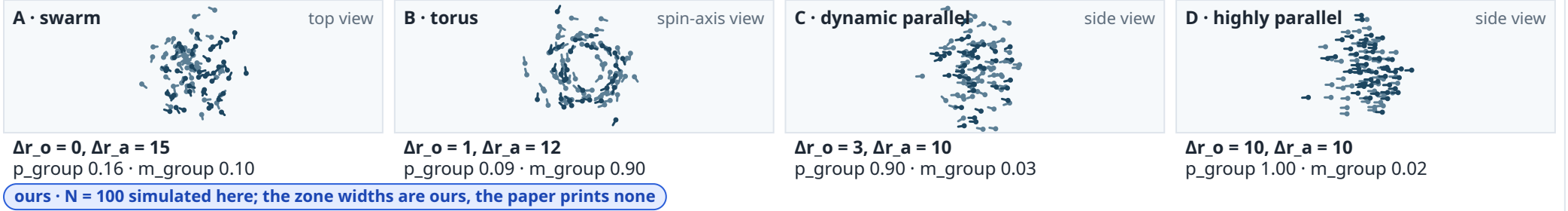


How a school swarms

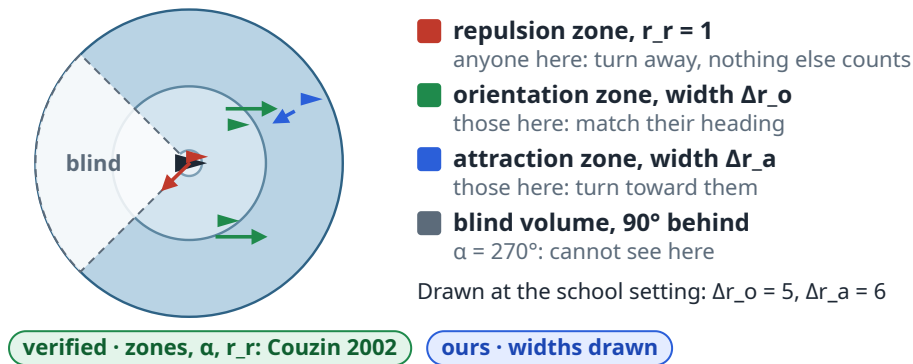
collective motion for the aquarium level's ten followers · the Couzin zone model, measured against our own Forth implementation

Lengths are in **body lengths** (BL; a fish is 3.5 in). The tank is $13.4 \times 3.4 \times 4.7$ BL inside. A **school** is polarised (p_group near 1); a **swarm** is not (p near 0.1): moving one zone width, Δr_o , switches between them.

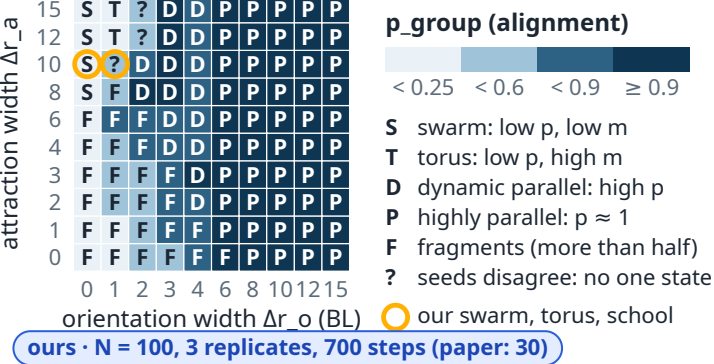
A · Four collective states from one rule, simulated here (N = 100, the paper's r_r , α , θ , s , σ)



B · The model: three zones and a blind volume



C · The map: p_group over the two zone widths



D · The rule in eight lines, and the Forth word that does each

- | | | |
|---|--|--|
| 1 | See the others, except behind | $u_{ij} \cdot v_i \geq \cos(\alpha/2)$ |
| 2 | Repulsion if anyone is within r_r | $d = -\sum u_{ij}$ (nothing else counts) |
| 3 | Else orientation and attraction | $d_o = \sum v_j, d_a = \sum u_{ij}; d = (\hat{d}_o + \hat{d}_a)/2$ |
| 4 | Nobody in any zone: keep the heading | $d = v_i$ |
| 5 | Turn toward d by at most $\theta\tau$, move $s\tau$ | $v \leftarrow \text{rotate}(v, d, \theta\tau); c \leftarrow c + s\tau v$ |
| 6 | The leader counts w times in sums 3 | $\sum w_j v_j, \sum w_j u_{ij}$ |
| 7 | Walls repel like a neighbour | wall within 0.6 BL : one more repulsion |
| 8 | A dart startles the followers nearby | swim away from the leader for 0.5 s |

sch-pair
sch-pair
sch-pair, sch-want
sch-want
sch-turn, sch-follow
sch-pair (weight)
sch-wall
sch-startle-all

The paper's random error ($\sigma = 0.05$ rad) is left out of the Forth: the caller adds it. Our units differ from the paper's, on purpose: at its $s = 3$ BL/s and $\theta = 40^\circ/s$ a 90° turn takes 6.75 BL of swimming (radius 4.3 BL), wider than the tank is deep (3.4 BL), and the tank runs below could not keep the fish in. Ours: $s = 2$ BL/s, $\theta = 120^\circ/s$, 1.50 BL, radius 0.95 BL.

verified · rules 1-5 and the paper's s , θ : Couzin 2002, eqns 1-3 and Fig. 3 ours · rules 6-8, our s , θ , the 0.6 BL wall, the 0.5 s startle

F · The Forth core, school.fth: the four rule words, and what they cost

sch-pair	168 B · 24 lines	sch-want	88 B · 11 lines	sch-turn	104 B · 12 lines	sch-follow	131 B · 17 lines
one neighbour: blind volume, then repulsion, orientation or attraction : sch-pair (j --) MB_YOU sc! MB_X MB_U you-delta MB_Y MB_U 1 + you-delta MB_Z MB_U 2 + you-delta MB_U vlen2 dup 0.000001 > over reach2 < & if sch-sqrt dup MB_DIST sc! 1 swap / MB_U vscale heading-dot-u MB_COSB par@ >= if r-rep inside if -1 MB_SUM_R acc+u MB_N_R count+ else r-ori inside if you-weight MB_SUM_0 acc+v MB_N_0 count+ else you-weight MB_SUM_A acc+u MB_N_A count+ then then then else drop then ;	the wanted direction from the three sums, or the old heading : sch-want MB_WANT vzero MB_N_R seen? if MB_SUM_R add-wanted else MB_N_0 seen? if MB_SUM_0 add-wanted then MB_N_A seen? if MB_SUM_A add-wanted then then MB_WANT vnorm 0.000001 > not if MB_WANT heading>sc then ;	turn toward it by at most $\theta\tau$, keep the heading unit length : sch-turn MB_VX me sch@ MB_WANT sc@ * MB_VY me sch@ MB_WANT 1 + sc@ * + MB_VZ me sch@ MB_WANT 2 + sc@ * + dup MB_COST par@ >= if drop else perp-all MB_PERP vnorm 0.000001 > not if perp-fallback then turn-all MB_WANT vnorm drop then ;	one follower; start to finish: scan, walls, startle, turn, move : sch-follow MB_ME sc! clear-sums sch-n 0 do i me <> if i sch-pair then loop sch-walls sch-want sch-startle? if sch-startle-away sch-startle-tick then sch-turn MB_WANT sc@ MB_NVX me sch! MB_WANT 1 + sc@ MB_NVY me sch! MB_WANT 2 + sc@ MB_NVZ me sch! 0 advance 1 advance 2 advance clamp-all ;				

3216 B

of the 64 KB dictionary (4.9 %): 73 words,
202 code lines; 1e-03 off couzin.py in one
tick

7.5 → 11.3 ms

one tick, 11 fish, Chromecast HD: the bare interpreter, then in the engine (with the camera and sway)

4.2 $\rightarrow$ 0.28 μ s

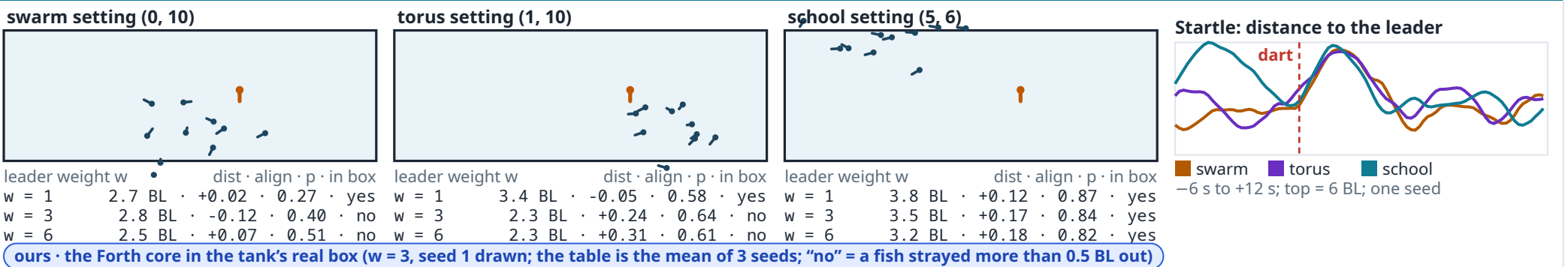
per mailbox call: three debug streams the engine ran on every call, found by this timing; 20 fps before

59.9 fps

the real level, ten followers, Chromecast
HD: p90 33.4 ms, Director 6.4 ms a tick

ours · our measurements: the bare interpreter, and a bench level (the Director runs school.fth every tick) on the real Chromecast, release build

G · The Forth, running in the tank: swarm, torus, school, and a startle



Sources · live links in the PDF; what could not be verified is marked

S1 [Couzin et al. 2002, J. Theor. Biol.](#)
verified zone model, states, metrics,
Fig. 3

S2 [Couzin et al. 2005, Nature](#)
unverified informed minority guides a group

S3 Pitcher 1983, via Wikipedia
unverified shoal to school: a continuum

S4 [Ocellaris clownfish \(Florida Museum\)](#)
unverified site-attached: not schooling
fish

S5 the aquarium level and this repository
 ours tank, fish, zForth, our runs

How to read the chips

verified the source was opened; the number is on its page
unverified from a summary, or not opened
ours our own choice, maths or measurement

Not claimed: that real clownfish school (they do not), that our sweep shows the paper's hysteresis (it is too coarse), or that the school is in the game: it is not wired in yet, only timed there. Data dated 2026-10-01.